

# FATING HONG

King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia

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## INTRODUCTION

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I am a postdoctoral fellow at the King Abdullah University of Science and Technology (KAUST), working with Prof. Peter Wonka. I received my Ph.D. in Computer Science and Engineering from the Hong Kong University of Science and Technology (HKUST), advised by Prof. Dan Xu. My research focuses on human-centric video generation, talking head synthesis, digital avatars and 4D human animation, and human motion generation and editing.

## RESEARCH INTEREST

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- Human-centric Video Generation / Talking Head Synthesis
- Digital Avatars and 4D Human Animation
- Human Motion Generation/Editing
- Audio-Visual Learning

## EDUCATION

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### Hong Kong University of Science and Technology

*Aug. 2021 - May. 2026*

- Ph.D. in Computer Science and Engineering
- Supervisor: Prof. Dan Xu

### Sun Yat-sen University

*Sep. 2018 - Jul. 2021*

- M.Sc. in Computer Science and Technology
- Supervisor: Prof. Wei-Shi Zheng

### South China University of Technology

*Aug. 2014 - Jul. 2018*

- B.Sc in Computer Science and Technology
- Supervisor: Prof. Sheng Bi

## EXPERIENCE

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### Center of Excellence for Generative AI, KAUST, Thuwal

*Aug. 2026 - Present*

- Postdoctoral Fellow
- Mentor: Prof. Peter Wonka

### Reality Labs Research, Meta, Zurich

*Jun. 2025 - Dec. 2025*

- Research Intern, 4D Avatars and Hair Modeling
- Mentor: Dr. Aljaz Bozic

### Hunyuan Lab, Tencent, Shenzhen

*Oct. 2024 - Mar. 2025*

- Research Intern, Human Video Generation (ACTalker, ICCV 2025)
- Mentor: Qinglin Lu

### Media Intelligence Lab, Adobe Research, San Jose

*May 2024 - Aug. 2024*

- Research Intern, Free-viewpoint Human Animation (CVPR 2025 Highlight)
- Mentors: Dr. Zhan Xu, Dr. Yang Zhou, Dr. Zhixin Shu, and Dr. Duygu Ceylan

- Depth-aware Talking Head Generation (DaGAN, CVPR 2022)

## GRANTS

### CAAI-Bose Quantum Computing Application Innovation Fund

2026

- Project: Research on 4D Gaussian World Models and Embodied Navigation via Quantum Optimization
- Role: Principal Investigator (joint with Jingke Meng, Sun Yat-sen University)

## PREPRINT

- **Fa-Ting Hong**, Peter Wonka<sup>✉</sup>, “SuperMotion: Source-Preserving Denoising for Text-Driven Human Motion Editing”, *arXiv:2610.01517 (Under review 2026)*.
- **Fa-Ting Hong**, Runzhen Liu, Luchuan Song, Hongmin Cai, Chuhua Xian<sup>✉</sup>, “EfficientSync: Real-Time Lip Synchronization via Deformation-Based Reference Texture Mixing”, *arXiv:2608.18832 (Under review 2026)*.
- **Fa-Ting Hong**, Dan Xu, “Arbitrary-Resolution 4D Human Animation via Continuous Feature Splatting”, *Under review 2026*.
- Hongzhe Liao, Chuhua Xian, Hongmin Cai, Haiyang Liu, **Fa-Ting Hong**<sup>✉</sup>, “SplitAvatar: One-shot Head Avatar with Autoregressive Gaussian Splitting”, *Under review 2026*.
- Runzhen Liu, Chuhua Xian, **Fa-Ting Hong**<sup>✉</sup>, “AtomicMotion: Learning Human Motion From Different Human Parts”, *Under review 2026*.
- **Fa-Ting Hong**, Yunfei Liu, Yu Li, Changyin Zhou, Fei Yu, Dan Xu, “DreamHead: Learning Spatial-Temporal Correspondence via Hierarchical Diffusion for Audio-driven Talking Head Synthesis”, *arXiv 2024*.
- **Fa-Ting Hong**, and Dan Xu, “Learning Online Scale Transformation for Talking Head Video Generation”, *arXiv 2024*.

## FIRST-AUTHOR & CORRESPONDING-AUTHOR PUBLICATIONS

- Runzhen Liu, Qinjie Lin, Yunfei Liu, Lijian Lin, Ye Zhu, Yu Li, Chuhua Xian, **Fa-Ting Hong**<sup>✉</sup>, “Identity-Preserving Video Dubbing Using Motion Warping”, *IJCV 2026*.
- Nan Lei, Yuan-Ming Li, Ling-An Zeng, Liang Xu, Zhi-Wei Xia, Hui-Wen Huang, **Fa-Ting Hong**<sup>✉</sup>, Wei-Shi Zheng, “PhysiGen: Integrating Collision-Aware Physical Constraints for High-Fidelity Human-Human Interaction Generation”, *ICASSP 2026*.
- **Fa-Ting Hong**, Zunnan Xu, Zixiang Zhou, Jun Zhou, Xiu Li, Qin Lin, Qinglin Lu, Dan Xu<sup>✉</sup>, “Audio-visual Controlled Video Diffusion with Masked Selective State Spaces Modeling for Natural Talking Head Generation”, *ICCV 2025*.
- **Fa-Ting Hong**, Zhan Xu, Haiyang Liu, Qinjie Lin, Luchuan Song, Zhixin Shu, Yang Zhou, Duygu Ceylan, Dan Xu<sup>✉</sup>, “Free-viewpoint Human Animation with Pose-correlated Reference Selection”, *CVPR 2025 (Highlight, 13.5% of accepted papers)*.
- **Fa-Ting Hong**, Li Shen, and Dan Xu<sup>✉</sup>, “DaGAN++: Depth-Aware Generative Adversarial Network for Talking Head Video Generation”, *IEEE TPAMI 2024 (Regular Paper)*.
- **Fa-Ting Hong**, and Dan Xu<sup>✉</sup>, “Implicit Identity Representation Conditioned Memory Compensation Network for Talking Head Video Generation”, *ICCV 2023*.
- **Fa-Ting Hong**, Longhao Zhang, Li Shen, and Dan Xu<sup>✉</sup>, “Depth-Aware Generative Adversarial Network for Talking Head Video Generation”, *CVPR 2022*.
- **Fa-Ting Hong**, Jia-Chang Feng, Dan Xu, Ying Shan, and Wei-Shi Zheng, “Cross-modal Consensus Network for Weakly Supervised Temporal Action Localization”, *ACM MM 2021*.

- **Fa-Ting Hong**, Xuanteng Huang, Wei-Hong Li, and Wei-Shi Zheng, “MINI-Net: Multiple Instance Ranking Network for Video Highlight Detection”, *ECCV* 2020.
- **Fa-Ting Hong**, Wei-Hong Li, and Wei-Shi Zheng, “Learning to Detect Important People in Unlabelled Images for Semi-supervised Important People Detection”, *CVPR* 2020.
- Wei-Hong Li\*, **Fa-Ting Hong\***, and Wei-Shi Zheng, “Learning to Learn Relation for Important People Detection in Still Images”, *CVPR* 2019 (\*Equal first author).

## CO-AUTHORED PUBLICATIONS

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- Haiyang Liu, Zhan Xu, **Fa-Ting Hong**, Hsin-Ping Huang, Yi Zhou, Yang Zhou, “Video Motion Graphs”, *ICCV* 2025.
- Shuling Zhao, **Fa-Ting Hong**, Xianshui Huang, Dan Xu, “Synergizing Motion and Appearance: Multi-Scale Compensatory Codebooks for Talking Head Video Generation”, *CVPR* 2025.
- Zunnan Xu, Zhentao Yu, Zixiang Zhou, Jun Zhou, Xiaoyu Jin, **Fa-Ting Hong**, Xiaozhong Ji, Junwei Zhu, Chengfei Cai, Shiyu Tang, Qin Lin, Xiu Li, Qinglin Lu, “HunyuanPortrait: Implicit Condition Control for Enhanced Portrait Animation”, *CVPR* 2025.
- Jun Zhou, Jiahao Li, Zunnan Xu, Hanhui Li, Yiji Cheng, **Fa-Ting Hong**, Qin Lin, Qinglin Lu, Xiaodan Liang, “FireEdit: Fine-grained Instruction-based Image Editing via Region-aware Vision Language Model”, *CVPR* 2025.
- Yu Wang, Yunfei Liu, **Fa-Ting Hong**, Meng Cao, Lijian Lin, Yu Li, “AnyTalk: Multi-modal Driven Multi-domain Talking Head Generation”, *AAAI* 2025.
- Jia-Run Du, Jia-Chang Feng, Kun-Yu Lin, **Fa-Ting Hong**, Xiao-Ming Wu, Zhongang Qi, Ying Shan, Wei-Shi Zheng, “Weakly-Supervised Temporal Action Localization by Progressive Complementary Learning”, *IEEE TCSVT* 2024.
- Yu-Kun Qiu, **Fa-Ting Hong**, Wei-Hong Li, and Wei-Shi Zheng, “Learning Relation Models to Detect Important People in Still Images”, *IEEE TMM* 2022.
- Jia-Chang Feng, **Fa-Ting Hong**, and Wei-Shi Zheng, “MIST: Multiple Instance Self-Training Framework for Video Anomaly Detection”, *CVPR* 2021.
- Ling-An Zeng, **Fa-Ting Hong**, Wei-Shi Zheng, Qi-Zhi Yu, Wei Zeng, Yao-Wei Wang, and Jian-Huang Lai, “Hybrid Dynamic-static Context-aware Attention Network for Action Assessment in Long Videos”, *ACM MM* 2020.
- Yuhong Liang, **Fa-Ting Hong**, Qinjie Lin, Sheng Bi, and Liqian Feng, “Optimization of Robot Path Planning Parameters Based on Genetic Algorithm”, *RCAR* 2017.

## PATENTS

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### Pose Correlation Based Digital Video Generation

- Inventor: Zhixin Shu, Zhan Xu, Yang Zhou, **Fa-Ting Hong**, Duygu Ceylan
- Docket No.: P13394-US
- Status: Filed
- Country/Region: US

### An image/video important people detection method combining deep learning and relationship modeling

- Inventor: Wei-Shi Zheng, **Fa-Ting Hong**
- Authorization Publication Number: CN111008558B
- Status: Authorized
- Country/Region: China

### Training methods, training devices and detection methods for vehicle trajectory detection models

- Inventor: Yunteng Xian, **Fa-Ting Hong**, Xianxu Xie, Ji Tang
- Application Publication Number: CN114612868A
- Status: Filed
- Country/Region: China

## AWARDS

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- CVPR 2025 Highlight (13.5% of accepted papers), Free-viewpoint Human Animation
- National Scholarship for Graduate Students, Ministry of Education of China, 2020
- National Scholarship (1/80), Ministry of Education of China, 2017

## ACADEMIC SERVICES

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- Reviewer: IEEE TPAMI, IJCV, IEEE TMM, CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH Asia, ACM MM

## MORE DETAILS

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For more information, please visit my personal page at: <https://harlanhong.github.io>.

## JUSTIFICATION ON THE QUALITY OF PUBLICATION VENUES

My dissertation chapters appear in the leading venues of computer vision. Almost all of them are rated A (the highest tier) by the China Computer Federation (CCF) list of recommended venues, which is widely used to assess research output in computer science. Acceptance rates below are approximate figures reported by the organizers.

| Venue                                                                           | Standing in the field                                                                                                                                                                                                           | Papers                                                                            |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| CVPR<br>(IEEE/CVF<br>Conf. on<br>Computer Vision<br>and Pattern<br>Recognition) | Flagship conference of computer vision. CCF-A. Among the highest-ranked publication venues of any discipline in Google Scholar Metrics. Acceptance about 22% (2025), 25% (2022). Highlights are about 13.5% of accepted papers. | 2022, 2025<br>(Highlight), plus<br>2019, 2020 and 3<br>co-authored 2025<br>papers |
| ICCV<br>(IEEE/CVF Int.<br>Conf. on<br>Computer Vision)                          | Premier biennial vision conference. CCF-A. Acceptance about 26% (2023), 24% (2025).                                                                                                                                             | 2023, 2025, plus 1<br>co-authored 2025<br>paper                                   |
| ECCV (European<br>Conf. on<br>Computer Vision)                                  | Premier biennial vision conference. CCF-A. Acceptance about 27% (2020).                                                                                                                                                         | 2020                                                                              |
| IEEE TPAMI                                                                      | Top journal in computer vision and machine intelligence. CCF-A, JCR Q1, impact factor above 20.                                                                                                                                 | 2024 (regular paper)                                                              |
| IJCV (Int.<br>Journal of<br>Computer Vision)                                    | Leading computer vision journal. CCF-A, JCR Q1.                                                                                                                                                                                 | 2026 (corresponding<br>author)                                                    |
| ACM Multimedia                                                                  | Top multimedia conference. CCF-A.                                                                                                                                                                                               | 2021, 2020                                                                        |
| AAAI                                                                            | Top artificial intelligence conference. CCF-A.                                                                                                                                                                                  | 2025                                                                              |
| IEEE TMM,<br>IEEE TCSVT                                                         | Leading multimedia and video journals. CCF-B, JCR Q1.                                                                                                                                                                           | 2022, 2024                                                                        |
| ICASSP                                                                          | Flagship signal processing conference. CCF-B.                                                                                                                                                                                   | 2026                                                                              |